Introduction to Economics

Class 11 — Production and costs

Pavel Kocourek

Universitat de Barcelona · Group A6

29 October 2026

Plan for today

Warm-up
Electricity for nothing?
What counts as a cost; in pairs: a yoga studio
One press, more printers
Total, average and marginal cost
In pairs: an ice-cream maker's costs
Break
Electricity for nothing: why
The U-shaped cost curves; in pairs
Short run and long run: one press or two
Economies of scale, and why they end
In pairs: a laundry grows
In one line; before Monday 2 November

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?

2. A friend offers 40 for the ticket. Now?

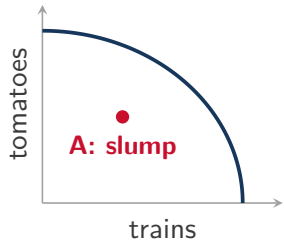
3. The concert is worth 60 to you. Go?

In pairs, two minutes; then hands per item.

Warm-up: what moves the frontier?

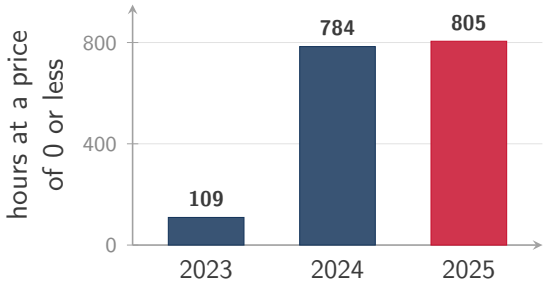
A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker
2. Workers arrive who can do either job
3. Factories idle in a slump reopen



In pairs, one minute; then hands.

Spain, 2025: about 800 hours of electricity at a price of zero or less



Almost all in the middle of the day, most in spring.

Why would anyone sell electricity for nothing?

Own count from the day-ahead prices of OMIE (the Spanish–Portuguese wholesale electricity market), mainland Spain (set per quarter-hour since October 2025: four of them count as one hour).

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700
= economic profit	**1,700**

Accountants count the money paid out; economists count every cost given up.

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

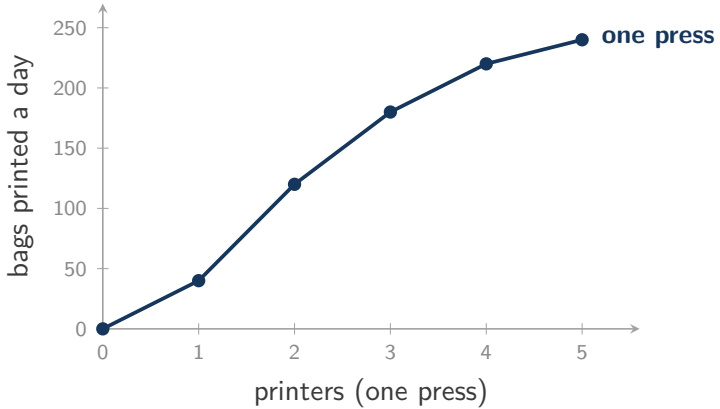
sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
2. Implicit costs?
3. Economic profit?
4. Economic profit 0: is she losing money?

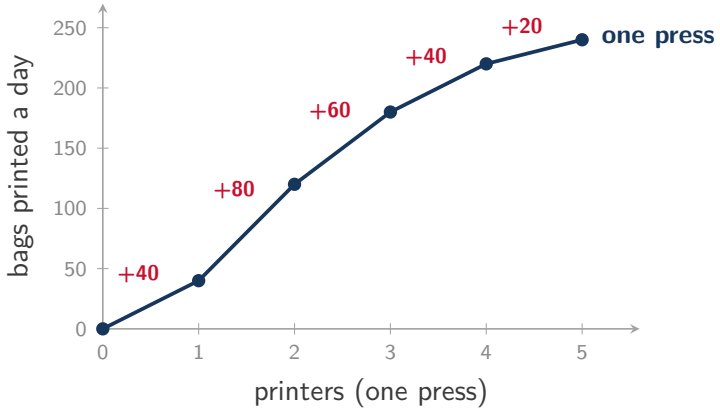
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.



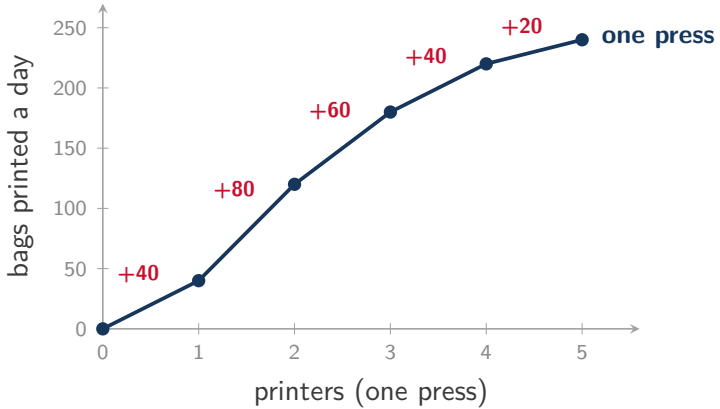
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.



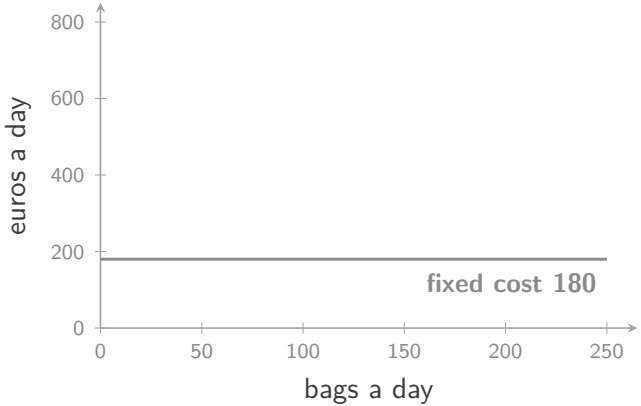
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.

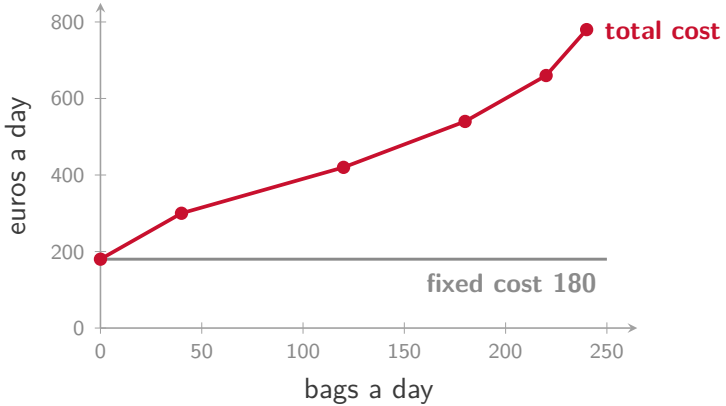


Marginal product: the bags one more printer adds. It falls once they crowd the press.

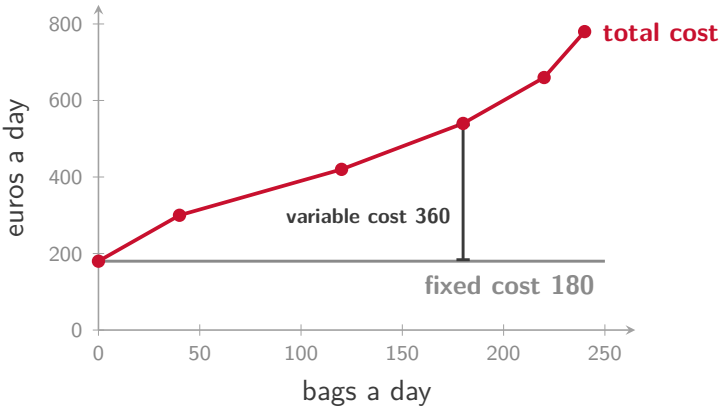
Total cost climbs faster as the press gets crowded



Total cost climbs faster as the press gets crowded



Total cost climbs faster as the press gets crowded



From the third printer on, each adds 120 but fewer bags: the curve gets steeper.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC
0	0	180
1	40	300
2	120	420
3	180	540
4	220	660
5	240	780

Fixed cost 180 a day · each printer 120 a day.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	**AFC** FC ÷ bags	**AVC** VC ÷ bags	**ATC** TC ÷ bags
0	0	180	—	—	—
1	40	300	4.50	3.00	7.50
2	120	420	1.50	2.00	3.50
3	180	540	1.00	2.00	**3.00**
4	220	660	0.82	2.18	**3.00**
5	240	780	0.75	2.50	3.25

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	**AFC** FC ÷ bags	**AVC** VC ÷ bags	**ATC** TC ÷ bags	**MC** extra cost ÷ extra bags
0	0	180	—	—	—	
1	40	300	4.50	3.00	7.50	3.00
2	120	420	1.50	2.00	3.50	1.50
3	180	540	1.00	2.00	**3.00**	2.00
4	220	660	0.82	2.18	**3.00**	3.00
5	240	780	0.75	2.50	3.25	6.00

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

MC = 120 ÷ the extra bags: it rises as each printer adds fewer.

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
2. ATC at 30 tubs?
3. MC of the tubs from 30 to 40?
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
2. At 6 euros?
3. A fourth farm sells 100 at either price. At 6?

Break

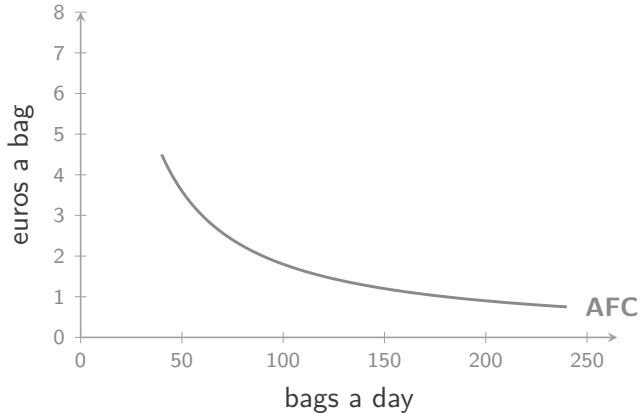
Ten minutes

Electricity for nothing: why?

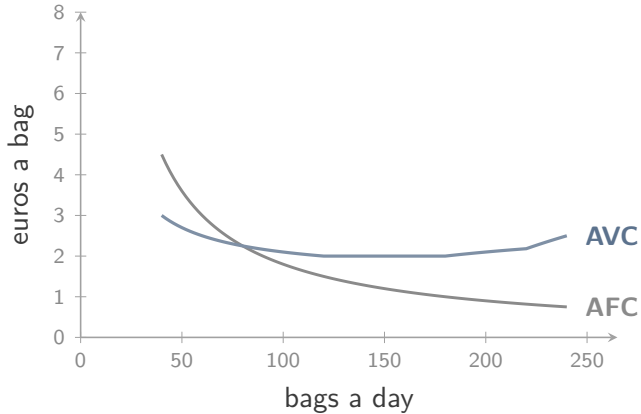
Revealed in class

Think back to slides 9 and 10: which of a solar farm's costs change if it switches off for an afternoon?

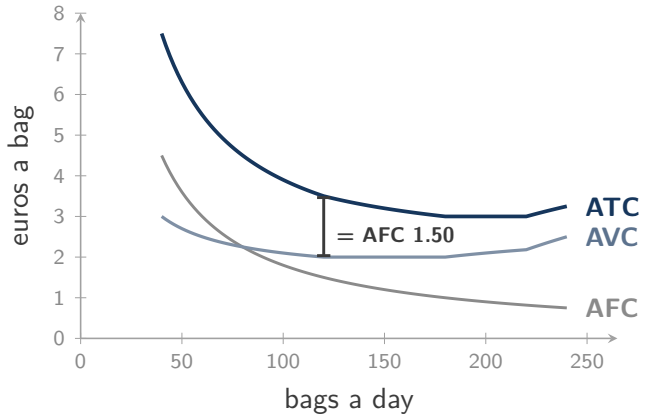
Fixed cost spread thinner, a crowded press: average total cost is a U



Fixed cost spread thinner, a crowded press: average total cost is a U

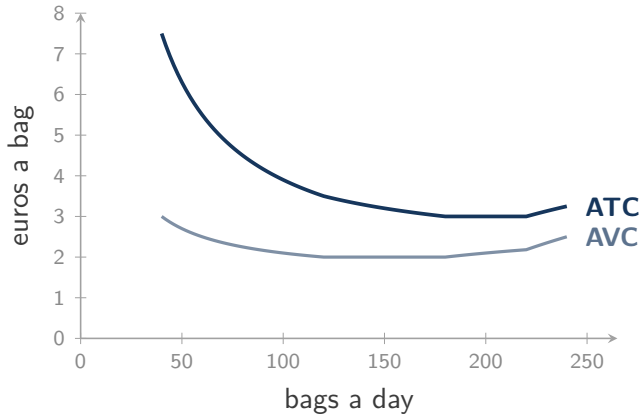


Fixed cost spread thinner, a crowded press: average total cost is a U

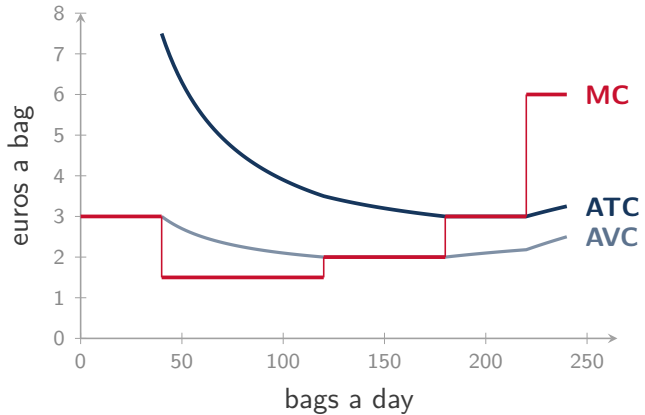


ATC = AFC + AVC: falling fixed cost per bag, then a crowded press.

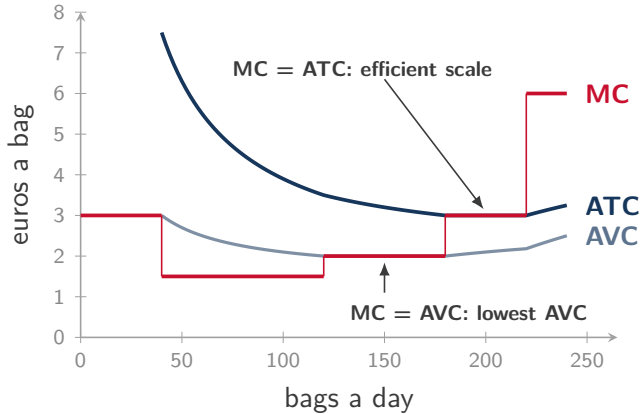
Marginal cost cuts through both averages at their lowest point



Marginal cost cuts through both averages at their lowest point



Marginal cost cuts through both averages at their lowest point



MC below the average pulls it down; above, pushes it up.

In pairs, two minutes: averages and the next one

Below the average or above it?

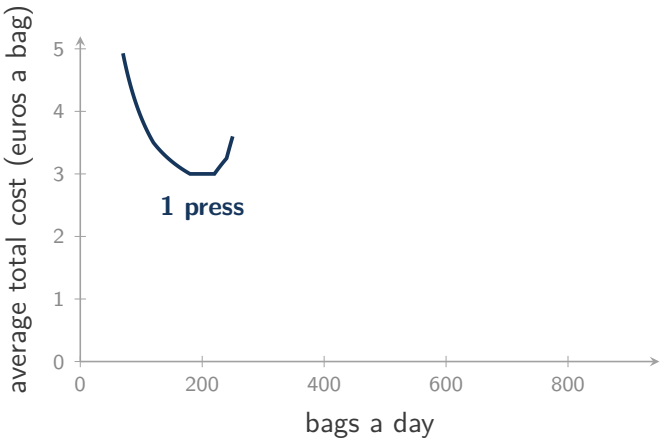
1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
3. Where MC equals ATC, ATC is...

Then hands per item.

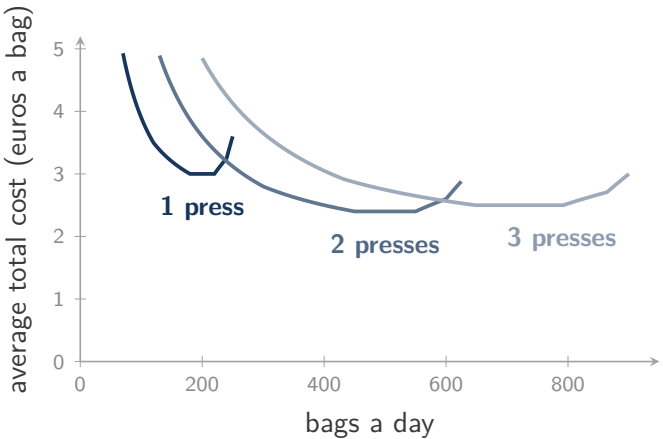
In the long run every cost can change: the shop can rent a second press

- Short run: the press is fixed; only printers change
- Long run: a second press, a bigger workshop
- In the long run every cost is variable
- How long? A print shop: months; a car plant: years

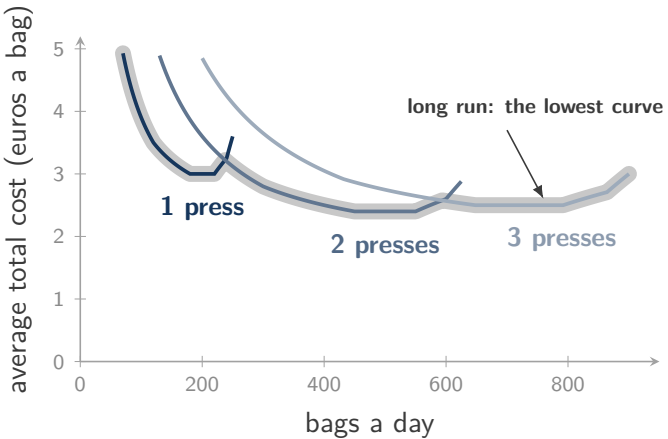
The long-run cost curve: the cheapest size for each output



The long-run cost curve: the cheapest size for each output



The long-run cost curve: the cheapest size for each output



In the long run the shop picks its size: the lowest curve at each output.

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**
330 bags	4.00	**diseconomies of scale**

What counts: does output grow faster than cost?

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Too big, dearer

more managers, slower decisions
messages lost on the way
workers far from the big picture

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
2. Doubled: ATC? What kind of returns?
3. Doubled again: ATC? Returns?
4. A hotel adds 100 kilos a day from next week. What can change by then?

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
2. The charge for a plastic bag doubles
3. Demand rises and totes get dearer: does supply shift too?

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."

2. "Electricity should be cheap for everyone."

3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?

A cost is what the firm gives up, paid out or not.

Once workers crowd the press, each adds less,

so marginal cost rises; it cuts average cost at its lowest point.

In the long run every cost can change: economies of scale, then diseconomies.

Monday: how much a firm sells at a given price.

Before Monday 2 November

- Monday: perfect competition; monopoly begins
- Read Mankiw & Taylor: Ch. 5, pp. 123–137
- Problem set 2 (Block 3): due Monday 2 November, on paper, in class
- Problem set 3 (Block 4): handed out Monday; due Thursday 12 November
- Practise in Sextant: the units the course page lists

pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ